

(Please write your Enrollment Number)

Enrollment No. _____

End-Term Examination
(CBCS)(SUBJECTIVE TYPE)(OffLine)
Course Name:<B.TECH>, Semester:<2>
(December, 2024)

Subject Code: BCS 110	Subject: Programming in C Language (Reappear upto 2023 batch)
Time :3 Hours	Maximum Marks :60

Note:Q1 is compulsory. Attempt one question each from the Units I, II, III & IV.

Q1	Attempt all parts	(2.5*8 =20)	CO Mapping
	a) Convert 512 KB to bytes and explain the significance of such conversions in a computer system.		CO1
	b) What is the difference between a flowchart and an algorithm in the context of problem-solving? Provide one example of each.		CO1
	c) What is the result of the following expression, and why? int a = 6, b = 3; printf("%d", a / b);		CO2
	d) Write a conditional if-else statement to check whether a number is positive or negative, and return 1 for positive and -1 for negative.		CO2
	e) What is the correct way to declare and initialize a 2x2 array in C, and how do you access the element at position [1,1]?		CO3
	f) What is the output of the following C code snippet? int i = 0; while (i < 5) { printf("%d ", i); i++; }		CO3
	g) Write a program to demonstrate "Call by Reference" using pointers. Show how a function can modify the value of a variable passed as an argument.		CO4
	h) Define the relationship between arrays and pointers. How is an array passed to a function?		CO4
UNIT I			CO Mapping
Q2	Consider the following program snippet: int a = 5, b = 10; int result = (a + b) * 2; printf("Result: %d", result); Analyze the program at a micro-level. Discuss the role of the ALU, registers, and memory in executing this program. Explain how the CPU performs each step from fetching to execution.	(10)	CO1
Q3	Given the following C program for finding the largest of three numbers: int a, b, c; printf("Enter three numbers: "); scanf("%d %d %d", &a, &b, &c); if (a > b && a > c) printf("%d is the largest", a);	(10)	CO1

	<pre> else if (b > a && b > c) printf("%d is the largest", b); else printf("%d is the largest", c); </pre> Analyze the logic of this program and convert it into a flowchart and algorithm. Discuss any improvements or optimizations you would make to the program.		
UNIT II			CO Mapping
Q4	A program contains multiple conditions with if-else, switch-case, and goto statements. Critique the program structure and suggest the best practices for readability and efficiency. Write an optimized program to check if a number is prime.	(10)	CO2
Q5	Design a program that uses scanf to input a floating-point number, an integer, and a character. The program should output whether the integer is odd or even, the character is uppercase or lowercase, and the floating-point number is positive or negative.	(10)	CO2
UNIT III			CO Mapping
Q6	Given the following code that includes a combination of for and while loops, predict its output and explain the flow of execution step by step. <pre> int i = 0, j = 5; for (i = 0; i < 3; i++) { while (j > 0) { printf("%d ", j); j--; } } </pre>	(10)	CO3
Q7	Write a C program that uses the strlen() and strcpy() standard library functions to copy a string entered by the user into another string. The program should display both the original and copied strings, and explain the working of these library functions.	(10)	CO3
UNIT IV			CO Mapping
Q8	Given the following code, determine its output and explain the working of pointers with respect to arrays. Also, identify any issues in the program. <pre> #include <stdio.h> int main() { int arr[] = {10, 20, 30, 40, 50}; int *ptr = arr; ptr = ptr + 2; printf("%d ", *ptr); ptr--; printf("%d ", *ptr); return 0; } </pre>	(10)	CO4
Q9	Write a C program that uses unions to store and print the data of either a student (name, age, marks) or a teacher (name, age, salary). Ensure that the program handles memory efficiently by choosing the correct type (student/teacher) at runtime. Discuss the memory management differences between structures and unions.	(10)	CO4